[Chinese | English]

- Dongshan PI One Issue Feedback and Exchange Area https://github.com/DongshanPI/Dongshanpi-Docs/discussions/2

Dongshan Pi One-Development Board

> Dongshan Pi No. 1 development board is the smallest Linux development board jointly promoted by the original chip manufacturer Xingchen Technology. It uses SSD202D main control chip. The smallest development board interface only retains a few common interfaces, including LED light buttons, FPCRGB display interface and one channel. The TYPE C USBHOST interface also has a TypeC interface specially used for power supply and debugging. At the same time, the SD card slot is added on the back and a special burning interface is reserved.

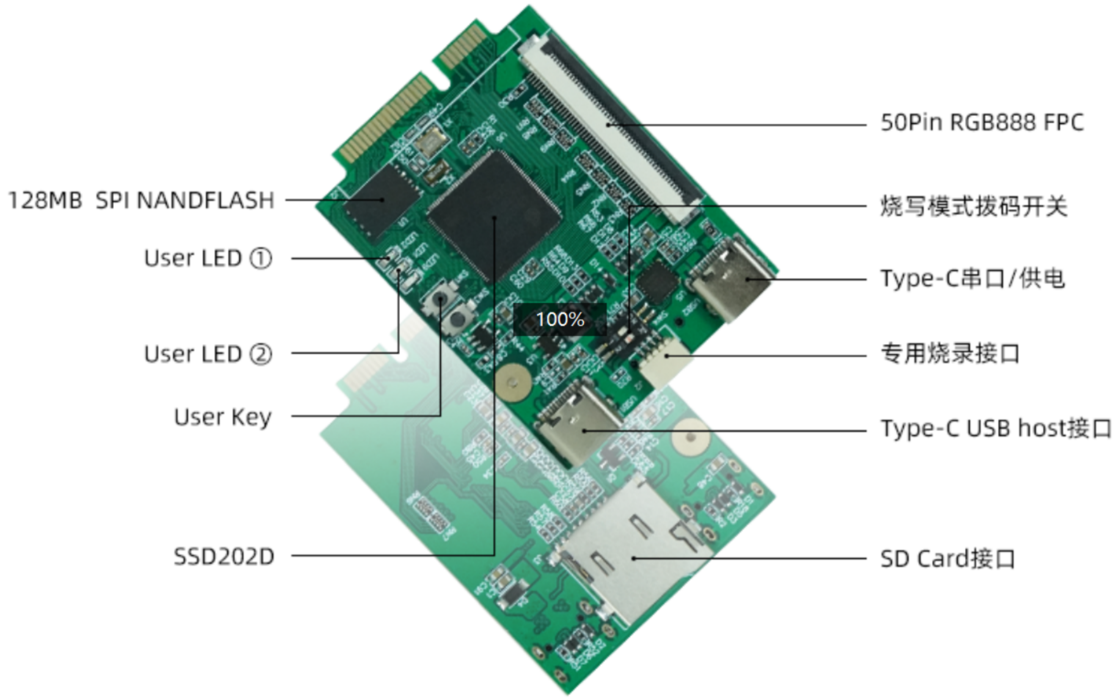
- Founded in 2017, SigmaStar is a technology company focusing on the field of artificial intelligence vision system chips. Its products are globally leading in many segments such as intelligent security, intelligent vehicle, and intelligent IoT.
- The founding team originated from the executive team of Morningstar Semiconductor (Mstar, the world's No. 1 in the field of display and TV chips, the largest chip supplier of Samsung, LG, TCL, CSOT, Konka, Hisense, BOE, etc., has merged with MediaTek).

hardware description

- Minimum board size

  The size of the development board refers to the standard MINI PCI-E specification, and on this basis, the chips and devices are adjusted accordingly. The approximate size is 3.8cm wide x 5.1cm long.

- Minimal Development Board Device Layout



- Minimum Development Board Specifications
- Main control chip: Star Technology SSD202D built-in 128MB DDR supports H264/H265 decoding and MJPG encoding.
- Storage: Onboard 128MB SPI NAND FLASH chip and dedicated SD card interface
- LED light: red x1 means pwr blue and green are user lights.
- Key: Hardware reset button x1 User button x1
- Display: 50Pin FPC RGB888 display output
- Power & Debug: The onboard dedicated USB to TTL chip supplies power to the entire board at the same time.
- usbHost: The USB HOST of the Type C interface supports connecting devices that support the USB protocol.
- Expansion interface: Use the MINI-PCI-E interface to connect to the backplane.
- Dongshan Pi One development board schematic file: https://cowtransfer.com/s/9b745ce262544c

Description of development resources

Supporting CPU Development Manual

The long-awaited Dongshan PI One (SSD202D) part of the commonly used peripheral CPU development manual has been released, which contains register addresses and descriptions.

Come and download and read.
Get connection: https://cowtransfer.com/s/0756a2af637745 which contains

- SSD202D_CHIPTOP_Baiwen.pdf
- SSD202D_CLKGEN_Baiwen.pdf
- SSD202D_EMAC_Baiwen.pdf
- SSD202D_GPIO_baiquest.pdf
- SSD202D_GPI_INT_Baiwen.pdf
- SSD202D_I2C_Baiwen.pdf
- SSD202D_INTR_CPUINT_Baiwen.pdf
- SSD202D_INTR_CTRL_Baiqing.pdf
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- SSD202D_PADTOP_Baiwen.pdf
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- SSD202D_RTC_Baiwen.pdf
- SSD202D_SAR_Baiqing.pdf
- SSD202D_SPI_Baiwen.pdf
- SSD202D_TIMER_Baiwen.pdf
- SSD202D_UART & FUART_Baiwen.pdf
- SSD202D_WDT_Baiwen.pdf

Chip original supporting SDK

> System version introduction, currently the whole set of SDK is adapted and explained based on the TAKOYAKI_DLC00V030 version released by the original Xingchen chip factory.

name	introduce
boot	Store the source code of the adapted uboot. For details, see Compile and Burn Boot on the left.
kernel	Store the source code of the adapted linux kernel. For details, see compiling and burning the kernel on the left.
project	It is mainly used to package image packages that can be upgraded. For details, please refer to the compilation and programming system on the left.

name	introduce
sdk	Store some simple demo code, see the application demo example on the left for details

- The advantages of the original chip SDK:

  This set of frameworks is an sdk specially adjusted by Xingchen Technology for product solutions, project plans, etc. It contains many demo examples in actual products. If we want to quickly implement products, we can use this set of sdk.

- Disadvantages of the original chip SDK

  Since this SDK is for production, a lot of framework changes have been made on this basis, which is different from our common Linux development. It needs to be used with the original development documents.

- For detailed acquisition and development methods, please refer to the **following** chapters on Getting Started with Original SDK Development.

Linux Community Edition SDK

> The community version is developed based on the mainline Linux Kernel Uboot and Buildroot.

- Community version advantages

  It is very convenient to learn the basic knowledge of Linux, familiar with the whole development process, and in line with the Linux development standards, there are abundant community reference books.

- Community Version Disadvantages

  Because there are many component drivers that are not used in the product and consume resources, it is not suitable for use as a product.

- For detailed acquisition and development methods, please refer to the **Community** Version Development section on the left.